

ASAHIKAWA, Japan

WMO: 474070

Lat: 43.7569N

Lon: 142.3722E

Elev: 460

StdP: 14.45

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.6	4.6	-3.1	4.7	1.3	0.6	5.8	5.8	22.8	26.1	19.6	24.4	2.4	80	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.4	85.8	72.9	82.9	70.6	80.5	69.1	75.0	82.6	73.3	80.0	71.6	77.5	8.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.6	122.9	79.4	71.0	116.3	77.6	69.4	110.1	76.1	38.9	82.8	37.3	80.5	35.8	77.7	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	17.3	14.7	DB	-6.3	90.6	3.9	2.7	-9.1	92.5	-11.4	94.1	-13.5	95.6	-16.4	97.5	(4)
(5)			WB	-6.7	76.9	3.9	2.1	-9.5	78.4	-11.7	79.6	-13.9	80.8	-16.7	82.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	18.4	20.4	29.1	41.8	54.5	62.9	70.3	71.0	62.2	49.3	36.4	24.1	(6)
(7)		DBStd	19.59	6.79	6.91	6.56	6.91	6.66	6.00	5.14	5.28	5.99	5.90	7.11	6.39	(7)
(8)		HDD50	4048	978	828	649	262	32	1	0	0	1	84	410	803	(8)
(9)		HDD65	7697	1443	1248	1114	695	330	106	15	12	123	485	858	1268	(9)
(10)		CDD50	2294	0	0	0	18	172	390	631	651	366	64	2	0	(10)
(11)		CDD65	466	0	0	0	0	4	44	181	198	39	0	0	0	(11)
(12)		CDH74	2923	0	0	0	3	101	350	1076	1194	198	1	0	0	(12)
(13)		CDH80	644	0	0	0	0	16	56	243	301	28	0	0	0	(13)
(14)	Wind	WSAvg	5.7	5.1	5.5	6.4	6.4	6.5	5.5	5.2	5.0	4.8	5.5	6.2	5.9	(14)
(15)	Precipitation	PrecAvg	43.0	2.8	2.2	2.2	2.1	2.6	2.8	4.4	6.0	5.3	4.3	4.6	3.9	(15)
(16)		PrecMax	56.5	5.3	4.5	4.4	4.7	5.3	5.2	13.7	18.8	11.7	8.4	8.9	6.5	(16)
(17)		PrecMin	28.8	0.8	0.7	0.6	0.4	0.5	0.4	0.9	0.9	1.0	1.5	2.6	1.9	(17)
(18)		PrecStd	6.5	1.0	0.9	1.0	0.9	1.0	1.1	2.4	3.6	2.7	1.5	1.3	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.4	41.8	52.1	72.4	81.9	84.8	88.9	89.9	83.7	70.2	60.7	44.1	(19)
(20)			MCWB	34.6	36.3	42.6	53.6	61.6	66.1	74.4	75.8	72.4	59.1	51.9	40.4	(20)
(21)		2%	DB	33.7	37.5	46.7	65.9	76.4	81.2	85.5	86.3	79.1	65.9	55.1	39.5	(21)
(22)			MCWB	31.0	33.4	38.7	51.4	58.7	65.7	72.8	74.0	68.3	56.9	48.8	36.5	(22)
(23)		5%	DB	31.0	34.5	43.0	59.8	72.3	78.1	82.6	83.4	75.7	62.8	51.3	36.2	(23)
(24)			MCWB	28.6	31.2	36.8	48.2	57.0	64.2	70.9	72.3	66.5	54.8	46.2	33.6	(24)
(25)		10%	DB	28.6	31.7	39.8	54.9	67.8	74.9	80.0	80.5	72.5	59.8	47.7	33.5	(25)
(26)			MCWB	26.3	28.8	34.7	45.1	55.1	63.0	69.7	70.6	64.4	53.4	43.7	31.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.9	37.5	43.6	56.3	63.7	70.1	76.9	77.4	73.5	61.5	53.6	40.5	(27)
(28)			MCD B	37.7	40.9	50.7	68.6	77.4	79.6	85.5	87.0	81.3	67.0	58.9	43.1	(28)
(29)		2%	WB	31.4	33.7	40.0	52.3	60.6	67.6	74.7	75.5	71.0	58.5	49.4	36.9	(29)
(30)			MCD B	33.5	36.8	45.1	63.0	72.6	77.0	81.8	83.6	76.6	63.5	54.1	39.1	(30)
(31)		5%	WB	28.8	31.3	37.2	49.1	58.4	65.9	73.0	73.9	68.4	56.4	46.5	33.8	(31)
(32)			MCD B	30.8	34.2	42.2	59.3	69.0	75.2	79.7	80.9	72.8	61.2	50.8	35.8	(32)
(33)		10%	WB	26.6	28.9	35.0	45.7	56.4	64.1	71.4	72.3	65.8	54.0	43.7	31.5	(33)
(34)			MCD B	28.4	31.6	39.3	53.8	65.9	72.7	77.4	78.4	71.1	59.0	47.4	33.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.5	15.5	15.2	17.9	20.3	18.4	16.2	15.4	16.8	16.9	12.2	11.3	(35)
(36)			MCDBR	14.3	17.2	19.3	27.9	28.5	24.5	19.8	18.8	19.8	21.2	19.7	14.4	(36)
(37)			MCWBR	12.4	14.1	12.9	15.0	13.5	11.3	9.1	8.5	10.1	12.8	14.0	12.1	(37)
(38)		5% WB	MCDBR	14.1	16.8	17.7	25.9	24.4	20.5	16.2	16.4	16.9	18.4	18.6	14.1	(38)
(39)			MCWBR	12.5	14.1	12.8	14.8	12.7	10.0	8.1	8.0	9.7	12.0	14.1	12.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.333	0.387	0.452	0.510	0.524	0.510	0.492	0.461	0.406	0.401	0.380	0.346	(40)
(41)		taud		1.968	1.817	1.728	1.781	1.870	1.991	2.108	2.212	2.315	2.181	2.108	1.982	(41)
(42)		Ebn at Noon		235	240	241	240	242	246	249	253	259	239	219	215	(42)
(43)		Edh at Noon		37	51	64	66	62	56	49	43	36	36	32	33	(43)
(44)	All-Sky Solar Radiation	RadAvg		429	693	1079	1412	1575	1594	1485	1360	1124	772	431	339	(44)
(45)		RadStd		49	80	104	130	115	157	175	89	87	53	44	34	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page